Image & Video Display Issues

🖼️ Image & Video Display Issues

This guide provides a systematic approach to resolving common visual artifacts, missing alarm snapshots, and streaming failures in the GCXONE platform.

❌ Missing Alarm Snapshots

If alarms are triggered but the Video Activity buffer shows empty image slots, verify the following:

1. Check Time Synchronization

Discrepancies between the device clock and GCXONE (even 5-10 seconds) are the **primary cause** of image retrieval failure.

- Ensure the device is using `time1.nxgen.cloud` or `pool.ntp.org`.
- **DST Check:** Ensure Daylight Saving Time is toggled **ON**. Without this, the system may fail during biannual time shifts.

2. Verify Recording Duration

Short recording buffers can cause retrieval misses.

- **Recommended Setting:** Configure your device to record at least **30 seconds** of video per event (not 10 seconds). This provides enough "padding" to capture three reliable snapshots even if clocks are slightly out of sync.

3. Storage Health (HDD Failure)

Check the device dashboard for **HDD Failure** or **Disk Full** alerts.

- If the hard drive is failing, the device will stop buffering video, and all subsequent alarms will have "Empty" image evidence.

🔁 Incorrect/Mixed Alarm Images

In rare cases, an alarm from "Site A" may display an image from "Site B." This is usually a symptom of **Central Cache Corruption**.

- **Scope Check:** Determine if the issue affects a single camera or the entire site.
- **Root Cause:** If images are routinely swapped across different physical NVRs, the local metadata index or the cloud cache index is likely out of sync.

- **Resolution:** Document the affected sites and escalate to the **Technical Systems Team** for a cache-flush operation.

📹 Streaming Failures (Local Mode)

If "Cloud Mode" works but "Local Mode" fails to load live video:

H.265 vs H.264

GCXONE cloud mode handles H.265 transcoding server-side. However, **Local Mode** sends raw packets directly from the Edge device to your browser.

1. **Check Encoding:** Verify if the camera is set to **H.265**.
2. **Install Codec:** Ensure your Windows operating system has the H.265 video extensions installed.
3. **Switch to H.264:** If the client machine cannot be updated, switch the camera's sub-stream encoding to **H.264** for maximum compatibility.

IP Camera Limitations

Independent IP cameras have limited processing power and often restrict the number of concurrent RTSP connections.

- If multiple operators are viewing the same IP camera, the device may refuse further connections.
- **Solution:** Switch the device to **Cloud Only Mode**. This routes all requests through a single cloud bridge, bypassing the device's connection limit.

💡 Quick Diagnostics Table

Symptom Probable Cause Immediate Action **Blue/Black Screen** H.265 Codec Missing Install extensions or switch to H.264. **"No Image" Button** Time Sync Drifting Re-sync device via NTP. **Site A shows Site B** Cache Corruption Escalate to Tech Team. **Laggy/Stuttering** Low Upload Bandwidth Reduce sub-stream bitrate/FPS.